

LECTURE #7

Econometrics I

DATA SCALING & FUNCTIONAL FORMS & MODEL COMPARISON

Jiri Kukacka, Ph.D.

Institute of Economic Studies, Faculty of Social Sciences, Charles University

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Midterm summary

- ▶ Aggregate results
- ▶ Summary statistics
- ▶ TOP mistakes

In the previous lecture #6

- ▶ We introduced testing multiple linear restrictions using F test:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(n - k - 1)} \sim F_{q, n-k-1}.$$

- ▶ Under MLR.1 through MLR.4, the **OLS estimator** $\hat{\beta}_j$ is **consistent** for all $j = 0, 1, \dots, k$.
- ▶ Even **MLR.4' Zero mean and zero correlation:**
 $\mathbb{E}(u) = 0$ and $\text{Cov}(x_j, u) = 0$ for $j = 1, 2, \dots, k$, is sufficient.
- ▶ Under MLR.1 through MLR.5, $\hat{\beta}_j$ is **asymptotically normally distributed** and **asymptotically efficient**.
- ▶ We discussed the Lagrange multiplier test: $LM = nR_U^2 \sim \chi_q^2$.
- ▶ Readings for lecture #7:
 - ▶ Chapter 6: 6.1–6.3.

Outline

Effects of data scaling on OLS statistics

More on functional form

- More on logarithmic forms

- More on models with quadratics

- Models with interaction terms

More on goodness-of-fit and model comparison

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Effects of data scaling on OLS statistics

Why do we rescale the data?

Mainly for reporting purposes:

- ▶ Too many decimals,
- ▶ Different units of measurement.

Effects of scaling (independent variable)

Instead of a standard population model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

consider a model ($c \neq 0$ being a constant)

$$y = \gamma_0 + \gamma_1(cx_1) + \gamma_2 x_2 + u.$$

Assuming MLR.1–MLR.6 hold, what is the relationship between:

- ▶ $\hat{\beta}_1$ and $\hat{\gamma}_1$, $\hat{\beta}_2$ and $\hat{\gamma}_2$?
- ▶ $se(\hat{\beta}_1)$ and $se(\hat{\gamma}_1)$?
- ▶ Relevant t statistics?
- ▶ Relevant p -values?
- ▶ Hint: use the definition $se(\hat{\beta}_j) = \frac{\hat{\sigma}}{\sqrt{SST_j(1-R_j^2)}}$.

Effects of scaling (independent variable)

- Estimate:

$$\hat{\beta}_1 = \hat{\gamma}_1 c \longrightarrow \boxed{\hat{\gamma}_1 = \frac{\hat{\beta}_1}{c}} \longrightarrow \hat{y} = \hat{\beta}_0 + \boxed{\frac{\hat{\beta}_1}{c}} cx_1 + \hat{\beta}_2 x_2.$$

- Standard error:

- $SST_{x_1} = \sum (x_{i1} - \bar{x}_1)^2,$

- $SST_{cx_1} = \sum (cx_{i1} - c\bar{x}_1)^2 = c^2 \sum (x_{i1} - \bar{x}_1)^2 = c^2 SST_{x_1},$

- $\boxed{se(\hat{\gamma}_1)} = \frac{\hat{\sigma}}{\sqrt{SST_{cx_1}(1-R_{x_1}^2)}} = \frac{\hat{\sigma}}{\sqrt{c^2 SST_{x_1}(1-R_{x_1}^2)}} =$

$$\frac{1}{|c|} \frac{\hat{\sigma}}{\sqrt{SST_{x_1}(1-R_{x_1}^2)}} = \boxed{\frac{se(\hat{\beta}_1)}{|c|}},$$

- t statistic: $\boxed{t_{\hat{\gamma}_1}} = \frac{\hat{\gamma}_1}{se(\hat{\gamma}_1)} = \frac{\hat{\beta}_1/c}{se(\hat{\beta}_1)/c} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} = \boxed{t_{\hat{\beta}_1}} \dots \text{if } c > 0,$
- p -value is based on the t stat., i.e., it does not change either.

Effects of scaling (dependent variable and logs)

What happens when:

- ▶ Dependent variable is rescaled?
- ▶ Dependent variable in the logarithmic form is rescaled?
- ▶ Independent variable in the logarithmic form is rescaled?

Effects of scaling (dependent variable)

Dependent variable rescaled:

$$\begin{aligned} cy &= \gamma_0 + \gamma_1 x_1 + \gamma_2 x_2 + cu \longrightarrow c\hat{y} = \boxed{c\hat{\beta}_0} + \boxed{c\hat{\beta}_1}x_1 + \boxed{c\hat{\beta}_2}x_2, \\ y &= \frac{\gamma_0}{c} + \gamma_1 \frac{x_1}{c} + \gamma_2 \frac{x_2}{c} + u. \end{aligned}$$

- ▶ Effect on **all parameters and estimates** is parallel to the independent variable rescaling, i.e., $\gamma_j = c\beta_j$ and $\hat{\gamma}_j = c\hat{\beta}_j$, $j = 0, 1, \dots, k$.
- ▶ $SST_{x_j/c}$ changes again:
 - ▶ $SST_{x_j/c} = \sum (x_{ij}/c - \bar{x}_j/c)^2 = \frac{1}{c^2} SST_{x_j}$,
 - ▶ $\boxed{se(\hat{\gamma}_j)} = \frac{\hat{\sigma}}{\sqrt{\frac{SST_{x_j}}{c^2}(1-R_{x_j}^2)}} = \boxed{|c| \cdot se(\hat{\beta}_j)}$.
- ▶ In the end, the estimate is rescaled by the factor of c , *se* by the factor of $|c|$, i.e., $|t \text{ stats}|$ and p -values remain unchanged.

Effects of scaling (logarithms)

- ▶ Dependent variable in logs rescaled:

$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

$$\log(cy) = \gamma_0 + \gamma_1 x_1 + \gamma_2 x_2 + v$$

$$\log(c) + \log(y) = \gamma_0 + \gamma_1 x_1 + \gamma_2 x_2 + v$$

$$\log(y) = \gamma_0 - \log(c) + \beta_1 x_1 + \beta_2 x_2 + u$$

$$\Rightarrow \boxed{\gamma_0 = \beta_0 + \log(c)}$$

$$\Rightarrow \widehat{\log(cy)} = \boxed{\hat{\beta}_0 + \log(c)} + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

$\Rightarrow$ **slope parameters/estimates are not affected**, only the intercept is adjusted.

- ▶ Independent variable in logs rescaled: parallel to the case above, only the intercept changes.
- ▶ **Rescaling within the logarithmic forms** (no matter if y or x_j or both) thus translates **only** into the **intercept**.

Illustrative example

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Source: Wooldridge (2012)

- ▶ Another type of 'scaling' (not mandatory): Standardized 'beta coefficients'

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More on logarithmic forms

- ▶ Consider the following model:

$$\log(y) = \beta_0 + \beta_1 \log(x_1) + \beta_2 x_2 + u.$$

- ▶ **Semi-elasticity** $100\hat{\beta}_2$ is only an **approximate percentage change** in $\hat{y}$, but it always lies between the absolute values for an increase and a decrease of x_2 .
- ▶ Exact change is: $\% \Delta \hat{y} = 100(\exp(\hat{\beta}_2 \Delta x_2) - 1)$.
- ▶ **Log-level form** is often used for relationships with an increasing rate of change (wages, sales, market value).
- ▶ **Level-log form** for a diminishing rate of change.
- ▶ Models with $\log(y)$ often more closely satisfy the CLM assumptions (linearity, homoskedasticity, normality).
- ▶ Logarithmic transformation also generally reduces the variation of variables and narrows its range (high sales, large salaries, sizeable populations).

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More on models with quadratics

- ▶ Polynomial forms are good approximations of various nonlinear relationships (Taylor expansions) and are useful for their interpretation.
- ▶ Consider the following model: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$.
- ▶ Partial derivative then implies:

$$\Delta \hat{y} = (\hat{\beta}_1 + 2\hat{\beta}_2 x) \Delta x.$$

- ▶ Effect of x on $\hat{y}$ is thus a function of the level of x dependent on values of $\hat{\beta}_1$ and $\hat{\beta}_2$.
- ▶ x^2 used for modelling a **diminishing effect** (of 'cumulative' variables): $\hat{\beta}_1 > 0$, $\hat{\beta}_2 < 0$, $\hat{\beta}_1 \gg |\hat{\beta}_2|$, $TP = -\hat{\beta}_1/(2\hat{\beta}_2)$.
- ▶ Higher polynomials can be utilized using the same logic (but the interpretation, of course, becomes a bit trickier).

Models with quadratics: Logarithms

- ▶ We may combine the quadratic relationship with the logarithmic one.
- ▶ Square of the logarithm must be used, not a square of x in the logarithm!
- ▶ Interpretation becomes somewhat more complicated, but the general idea of a quadratic relationship remains unchanged.
- ▶ Consider the following model:

$$\log(y) = \beta_0 + \beta_1 \log x + \beta_2 (\log(x))^2 + u.$$

- ▶ To get the relationship between x and y , we again use the partial derivative but now with respect to $\log x$:

$$\% \Delta \hat{y} = (\hat{\beta}_1 + 2\hat{\beta}_2 \log(x)) \% \Delta x.$$

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Models with interaction terms

- ▶ Sometimes, the effect of one explanatory variable on the dependent variable can itself be dependent on the magnitude of yet another explanatory variable.
- ▶ To make this less confusing, consider the following model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u.$$

- ▶ To see the effect of x_1 on Δy , we can again use the partial derivative to get

$$\Delta \hat{y} = (\hat{\beta}_1 + \hat{\beta}_3 x_2) \Delta x_1.$$

- ▶ This is called an **interaction effect**.
- ▶ Significance of β_3 , i.e., the existence of the interaction effect, can be easily tested using the standard procedures (t test, possibly F test).

Note on reporting

- ▶ As in these specific cases, the respective β s do not give us the effect straight away; it is useful to report the effects for some interesting values (average, median, quantiles) of the independent variable of interest.
- ▶ Connected to the previous point, reporting a range (min to max) of the effect is sometimes useful.
- ▶ The transformations you use should be reasonable and possible to interpret.

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Goodness-of-fit measures

- ▶ So far, we have considered the coefficient of determination R^2 .
- ▶ Adding an independent variable to a model never decreases R^2 , which makes it useless for variable selection/model construction.
- ▶ As an alternative goodness-of-fit measure, the adjusted R^2 labeled as $\bar{R}^2$ controls for the number of explanatory variables k :

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\frac{SSR}{n}}{\frac{SST}{n}},$$
$$\boxed{\bar{R}^2} = 1 - (1 - R^2) \frac{n-1}{n-k-1} = \boxed{1 - \frac{\frac{SSR}{n-k-1}}{\frac{SST}{n-1}}}.$$

- ▶ Final formula provides the logic behind $\bar{R}^2$, i.e., the interplay between SSR and df .
- ▶ Asymptotically, $\bar{R}^2 = R^2$.
- ▶ Interestingly, $\bar{R}^2$ can be even negative.

Adjusted $\bar{R}^2$

Adjusted coefficient of determination $\bar{R}^2$ has two interesting and potentially useful properties:

- ▶ When we add a new independent variable, $\bar{R}^2$ increases if and only if the t statistic on the new variable is greater than one in absolute value.
- ▶ When a group of new independent variables is added, $\bar{R}^2$ increases if and only if the F statistic on the group of new variables is greater than one.

Choosing between non-nested models

- ▶ Consider two models:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

$$y = \gamma_0 + \gamma_1 x_1 + \gamma_2 x_3 + v.$$

- ▶ These are called **non-nested models** as neither one is a special case of the other.
- ▶ We cannot use the F statistics for non-nested models (creating a broader composite model with all independent variables can lead to ambiguous results of hypotheses testing).
- ▶ R^2 can be used only for the same number of independent variables.
- ▶ $\bar{R}^2$ can also be used to compare various specifications (e.g., for the logarithmic vs. quadratic form of the independent variable).
- ▶ However, the **dependent variable** needs to have **the same functional form** across all competing models.

Seminars and the next lecture

- ▶ Seminars:
 - ▶ data scaling,
 - ▶ practicing logarithmic and quadratic functional forms,
 - ▶ practicing interaction terms,
 - ▶ model comparison.
- ▶ Next lecture #8:
 - ▶ selection of explanatory variables,
 - ▶ prediction and residual analysis,
 - ▶ multiple regression with qualitative information,
 - ▶ single binary/dummy independent variable,
 - ▶ using dummy variables for multiple categories.
- ▶ Readings for lecture #8:
 - ▶ Chapter 6: 6.4, Chapter 7: 7.1–7.3.

Appendix: Standardized 'beta coefficients' (not mand.)

- ▶ Sometimes, we are not interested in interpreting the effects in measurement units (usually due to non-standard measurement units of the variable of interest) but in **standard deviations**.
- ▶ We thus **standardize** the variables of interest to obtain the standardized '**beta coefficients**'.
- ▶ All variables are put on 'equal grounds'.
- ▶ Effects are interpreted as changes in standard deviations, often used in sociology and for survey analyses.
- ▶ Statistical significance **unaffected** (t stats, p -values).
- ▶ In a simple regression model, the beta coefficient b equals the correlation coefficient between the dependent and the independent variable.

Standardized 'beta coefficients': Derivation

- ▶ Starting from the estimated model

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \dots + \hat{u}_i,$$

we can subtract the corresponding averages from each variable in the equation to get

$$\hat{y}_i - \bar{y} = 0 + \hat{\beta}_1 (x_{1i} - \bar{x}_1) + \dots + (\hat{u}_i - 0).$$

- ▶ Having $\hat{\sigma}_y$ as a sample standard deviation of y and $\hat{\sigma}_j$, $i = 1, \dots, k$, as sample standard deviations of the independent variables, we can rewrite the equation as

$$\frac{y_i - \bar{y}}{\hat{\sigma}_y} = \hat{\beta}_1 \frac{\hat{\sigma}_1}{\hat{\sigma}_y} \frac{x_{1i} - \bar{x}_1}{\hat{\sigma}_1} + \dots + \frac{\hat{u}_i}{\hat{\sigma}_y}.$$

- ▶ Finally, using the z-score notation, we can write the equation as

$$z_y = \hat{b}_1 z_1 + \dots + \hat{e}_i,$$

giving us the **standardized beta coefficients** as

$$\hat{b}_j = \frac{\hat{\sigma}_j}{\hat{\sigma}_y} \hat{\beta}_j.$$